STUDENT PERFORMANCE ANALYSIS

Exploratory Data Analysis (EDA) Using Python

INTERNSHIP PROJECT REPORT

Internship Details:

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Report Title:

STUDENT PERFORMANCE ANALYSIS

ABSTRACT:

The Student Performance Analysis project focuses on exploring and analyzing students' academic performance using Python. The project uses the Student Performance dataset, which contains information about students' personal, social, and academic factors.

Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn are used to perform data analysis and visualization. Exploratory Data Analysis (EDA) techniques are applied to examine students' final grades, study time, absences, previous failures, internet access, family support, and other factors that may be related to academic performance.

Different graphs and statistical methods are used to identify patterns and relationships within the dataset. The analysis helps in understanding student performance and identifying factors that may influence academic results. Overall, this project provides practical experience in data handling, statistical analysis, data visualization, and interpretation using Python.

1. INTRODUCTION:

Student performance is an important factor in understanding the academic progress of students. Academic results can be influenced by several factors, such as study time, attendance, previous academic performance, family support, and students' personal and social conditions.

The Student Performance Analysis project uses the Student Performance dataset to explore students' academic information and understand the factors related to their final grades. The dataset contains information about students, including age, study time, absences, previous failures, and final academic grades.

In this project, Python is used to perform Exploratory Data Analysis (EDA). Libraries such as Pandas, NumPy, Matplotlib, and Seaborn are used to organize data, perform statistical analysis, and create visualizations. Different graphs and comparisons are used to identify patterns and relationships in student performance.

This analysis helps in understanding the dataset and gaining useful insights into the factors associated with students' academic results.

2. PROBLEM STATEMENT:

Student academic performance can vary due to several personal, social, and academic factors. Understanding these factors is important to identify patterns that may be associated with students' final grades.

The main problem addressed in this project is to explore the Student Performance dataset and understand how factors such as study time, absences, previous failures, family support, and other student-related attributes are associated with academic performance.

Therefore, this project uses Exploratory Data Analysis (EDA) techniques to examine the available data, compare different student attributes, and identify meaningful patterns through statistical analysis and visualizations.

3. OBJECTIVES OF THE PROJECT:

1.To understand and explore the Student Performance dataset.

2.To analyze students' academic performance using Python.

3.To study the relationship between study time and final grades.

4.To examine how absences and previous failures are related to academic performance.

5.To analyze the influence of personal, social, and family-related factors.

6.To use graphs and visualizations to identify patterns in the dataset.

7.To gain practical experience in data analysis and visualization using Python.

4. SCOPE OF THE PROJECT:

The scope of this project is to analyze students' academic performance using the Student Performance dataset. The project focuses on exploring different academic, personal, and social attributes of students and understanding their relationship with final grades.

The analysis includes studying factors such as study time, absences, previous failures, internet access, family support, school support, age, and other attributes available in the dataset.

The project uses Python libraries such as Pandas and NumPy for data handling and statistical analysis, while Matplotlib and Seaborn are used to

create visualizations. Different graphs and comparisons are used to identify patterns and understand the data.

This project is limited to Exploratory Data Analysis (EDA) and interpretation of the available dataset. It does not establish that any particular factor directly causes changes in students' academic performance.

5. TOOLS AND TECHNOLOGIES USED:

Programming Language:

- Python

Libraries Used:

1. **Pandas:** Used for loading, organizing, cleaning, and analyzing the dataset.
2. **NumPy:** Used for numerical operations and statistical calculations.
3. **Matplotlib:** Used to create graphs and visual representations of the data.
4. **Seaborn:** Used to create statistical visualizations and explore relationships between variables.

Development Environment:

- Jupyter Notebook

Dataset:

- Student Performance Dataset (`student-mat.csv`)

Fig 1: Importing the required Python libraries:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

print("All libraries imported successfully!")
```

```
All libraries imported successfully!
```

6. DATASET DESCRIPTION:

The Student Performance dataset is used in this project to analyze students' academic performance. The dataset contains information about students' personal, social, and academic characteristics.

The dataset used for this analysis is `student-mat.csv`. It contains 395 records and 33 columns. Each row represents a student, while each column represents a particular attribute.

Some of the important attributes include:

1. **School:** The school attended by the student.
2. **Sex:** Gender of the student.
3. **Age:** Age of the student.
4. **Address:** Whether the student lives in an urban or rural area.
5. **Study Time:** Weekly study time of the student.
6. **Absences:** Number of school absences.
7. **Previous Failures:** Number of previous class failures.
8. **Internet:** Availability of internet access at home.
9. **Family Support:** Support received from family.
10. **G1, G2 and G3:** First-period, second-period, and final grades.

The dataset helps explore student-related attributes and examine their association with academic performance.

7. METHODOLOGY:

The project follows an Exploratory Data Analysis (EDA) approach to understand the Student Performance dataset. The analysis is carried out using Python in Jupyter Notebook.

The following steps are followed in the project:

1. Dataset Loading: The Student Performance dataset (`student-mat.csv`) is loaded using the Pandas library.

2. Data Inspection: The dataset is examined to understand its structure, dimensions, column names, and data types.

3. Data Cleaning: Missing values and duplicate records are checked to understand the quality of the dataset.

4. Statistical Analysis: Descriptive statistics are used to understand the distribution of numerical attributes.

5. Data Visualization: Histograms, bar charts, scatter plots, box plots, pie charts, and heatmaps are used to visualize different attributes.

6. Relationship Analysis: Different student attributes are compared with final grades to identify patterns and associations.

7. Interpretation: The observations from statistical analysis and visualizations are interpreted to understand student performance.

8. DATA LOADING AND INITIAL INSPECTION:

The first step of the project is to load the Student Performance dataset into the Jupyter Notebook using the Pandas library.

The dataset is stored in CSV format and is loaded using the `read_csv()` function. After loading the dataset, initial inspection is performed to understand its structure and contents.

The dataset contains 395 rows and 33 columns. Each row represents a student, and each column represents a particular student attribute.

The following Pandas functions are used for initial inspection:

1. **`head()`** – Displays the first five rows of the dataset.
2. **`tail()`** – Displays the last five rows of the dataset.
3. **`shape`** – Displays the number of rows and columns.
4. **`info()`** – Provides information about column names, data types, and non-null values.
5. **`describe()`** – Displays statistical information about numerical columns.
6. **`isnull().sum()`** – Checks for missing values in each column.

These functions help in understanding the dataset before performing further analysis.

Fig 2.Loading the Student Performance dataset using Pandas:

```python
import zipfile
import pandas as pd
from io import BytesIO

with zipfile.ZipFile("student+performance.zip", "r") as outer_zip:
    inner_data = outer_zip.read("student.zip")

with zipfile.ZipFile(BytesIO(inner_data), "r") as inner_zip:
    with inner_zip.open("student-mat.csv") as csv_file:
        df = pd.read_csv(csv_file, sep=";")

print("Dataset loaded successfully!")
print("Dataset shape:", df.shape)
```

```
Dataset loaded successfully!
Dataset shape: (395, 33)
```

Fig 3: Displaying the first five rows of the Student Performance dataset.

```python
import pandas as pd
import zipfile
from io import BytesIO

with zipfile.ZipFile("student+performance.zip", "r") as outer_zip:
    inner_data = outer_zip.read("student.zip")

with zipfile.ZipFile(BytesIO(inner_data), "r") as inner_zip:
    with inner_zip.open("student-mat.csv") as csv_file:
        df = pd.read_csv(csv_file, sep=";")

print("Dataset loaded successfully!")
print("Dataset shape:", df.shape)
df.head()
```

```
Dataset loaded successfully!
Dataset shape: (395, 33)
```

	school	sex	age	address	famsize	Pstatus	Medu	Fedu	Mjob	Fjob	...	famrel	freetime	goout	Dalc	Walc	health	absences	G1	G2	G3
0	GP	F	18	U	GT3	A	4	4	at_home	teacher	...	4	3	4	1	1	3	6	5	6	6
1	GP	F	17	U	GT3	T	1	1	at_home	other	...	5	3	3	1	1	3	4	5	5	6
2	GP	F	15	U	LE3	T	1	1	at_home	other	...	4	3	2	2	3	3	10	7	8	10
3	GP	F	15	U	GT3	T	4	2	health	services	...	3	2	2	1	1	5	2	15	14	15
4	GP	F	16	U	GT3	T	3	3	other	other	...	4	3	2	1	2	5	4	6	10	10

5 rows × 33 columns

Fig 4 : Dataset dimensions, column names, and information.

```python
print("Number of students:", df.shape[0])
print("Number of columns:", df.shape[1])

print("\nColumn names:")
print(df.columns.tolist())

print("\nDataset information:")
df.info()
```

```
Number of students: 395
Number of columns: 33

Column names:
['school', 'sex', 'age', 'address', 'famsize', 'Pstatus', 'Medu', 'Fedu', 'Mjob', 'Fjob', 'reason', 'guardian', 'traveltime', 'studytime', 'failures',
'schoolsup', 'famsup', 'paid', 'activities', 'nursery', 'higher', 'internet', 'romantic', 'famrel', 'freetime', 'goout', 'Dalc', 'Walc', 'health', 'abs
ences', 'G1', 'G2', 'G3']

Dataset information:
<class 'pandas.DataFrame'>
RangeIndex: 395 entries, 0 to 394
Data columns (total 33 columns):
```

```
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   school      395 non-null    str
 1   sex         395 non-null    str
 2   age         395 non-null    int64
 3   address     395 non-null    str
 4   famsize     395 non-null    str
 5   Pstatus     395 non-null    str
 6   Medu        395 non-null    int64
 7   Fedu        395 non-null    int64
 8   Mjob        395 non-null    str
 9   Fjob        395 non-null    str
 10  reason      395 non-null    str
 11  guardian    395 non-null    str
 12  traveltime  395 non-null    int64
 13  studytime   395 non-null    int64
 14  failures    395 non-null    int64
 15  schoolsup   395 non-null    str
 16  famsup      395 non-null    str
 17  paid        395 non-null    str
 18  activities  395 non-null    str
 19  nursery     395 non-null    str
 20  higher      395 non-null    str
 21  internet    395 non-null    str
 22  romantic    395 non-null    str
 23  famrel      395 non-null    int64
 24  freetime    395 non-null    int64
 25  goout       395 non-null    int64
 26  Dalc        395 non-null    int64
 27  Walc        395 non-null    int64
 28  health      395 non-null    int64
 29  absences    395 non-null    int64
 30  G1          395 non-null    int64
 31  G2          395 non-null    int64
 32  G3          395 non-null    int64
dtypes: int64(16), str(17)
memory usage: 102.0 KB
```

9. DATA CLEANING AND PREPROCESSING:

Data cleaning and preprocessing are important steps in preparing a dataset for analysis. In this project, the Student Performance dataset is checked to understand its data quality before performing further analysis.

The following checks are performed:

1. **Missing Values:** The `isnull().sum()` function is used to identify missing values in each column.
2. **Duplicate Records:** The `duplicated().sum()` function is used to check whether duplicate rows are present in the dataset**.**
3. **Data Types:** The `info()` function is used to examine the data types of different columns.
4. **Data Consistency:** The dataset is inspected to understand whether the values are in a suitable format for analysis.

These preprocessing checks help assess the quality and structure of the dataset before visualization and statistical analysis.

Fig 5: Checking missing values in the Student Performance dataset:

```
print("MISSING VALUES IN EACH COLUMN")

print(df.isnull().sum())

print("\nTotal missing values:")
print(df.isnull().sum().sum())
```

```
MISSING VALUES IN EACH COLUMN
school        0
```

```
sex           0
age           0
address       0
famsize       0
Pstatus       0
Medu          0
Fedu          0
Mjob          0
Fjob          0
reason        0
guardian      0
traveltime    0
studytime     0
failures      0
schoolsup     0
famsup        0
paid          0
activities    0
nursery       0
higher        0
internet      0
romantic      0
famrel        0
freetime      0
goout         0
Dalc          0
Walc          0
health        0
absences      0
G1            0
G2            0
G3            0
dtype: int64

Total missing values:
0
```

Fig 6: Checking duplicate records in the dataset:

```
print("Total duplicate rows:")
print(df.duplicated().sum())
```

```
Total duplicate rows:
0
```

10. EXPLORATORY DATA ANALYSIS (EDA):

Exploratory Data Analysis (EDA) is the process of examining and understanding a dataset using statistical methods and visualizations. It helps identify patterns, trends, distributions, and relationships between different variables.

In this project, EDA is performed on the Student Performance dataset to explore students' academic results and the factors associated with them.

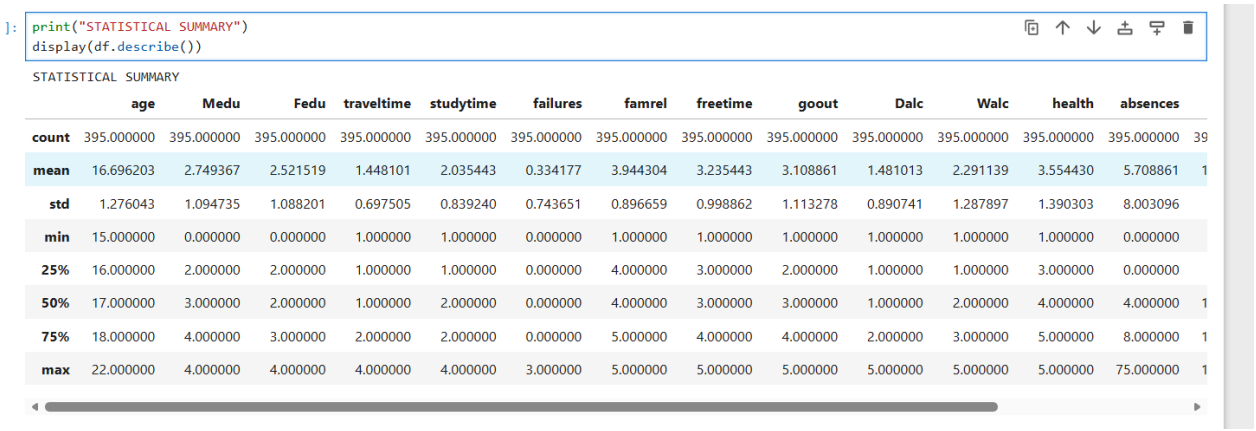
Different statistical methods and visualizations are used to study the dataset. The analysis focuses on final grades, study time, absences, previous failures, family support, internet access, and other student-related attributes.

The following techniques are used in this project:

1. Statistical summary of the dataset.
2. Analysis of students' final grades.
3. Distribution analysis using histograms.
4. Comparison of student attributes using bar charts.
5. Relationship analysis using scatter plots.
6. Comparison of numerical variables using box plots.
7. Correlation analysis using a heatmap.
8. Visualization of pass and fail categories.

These techniques help in understanding the dataset and identifying useful patterns related to student performance.

Fig 7: Statistical summary of the Student Performance dataset:



```
]: print("STATISTICAL SUMMARY")
   display(df.describe())
```

STATISTICAL SUMMARY

	age	Medu	Fedu	traveltime	studytime	failures	famrel	freetime	goout	Dalc	Walc	health	absences	
count	395.000000	395.000000	395.000000	395.000000	395.000000	395.000000	395.000000	395.000000	395.000000	395.000000	395.000000	395.000000	395.000000	39
mean	16.696203	2.749367	2.521519	1.448101	2.035443	0.334177	3.944304	3.235443	3.108861	1.481013	2.291139	3.554430	5.708861	1
std	1.276043	1.094735	1.088201	0.697505	0.839240	0.743651	0.896659	0.998862	1.113278	0.890741	1.287897	1.390303	8.003096	
min	15.000000	0.000000	0.000000	1.000000	1.000000	0.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	0.000000	
25%	16.000000	2.000000	2.000000	1.000000	1.000000	0.000000	4.000000	3.000000	2.000000	1.000000	1.000000	3.000000	0.000000	
50%	17.000000	3.000000	2.000000	1.000000	2.000000	0.000000	4.000000	3.000000	3.000000	1.000000	2.000000	4.000000	4.000000	1
75%	18.000000	4.000000	3.000000	2.000000	2.000000	0.000000	5.000000	4.000000	4.000000	2.000000	3.000000	5.000000	8.000000	1
max	22.000000	4.000000	4.000000	4.000000	4.000000	3.000000	5.000000	5.000000	5.000000	5.000000	5.000000	5.000000	75.000000	1

11. ANALYSIS OF STUDENTS' FINAL GRADES:

The final grade is an important measure of students' academic performance. In the Student Performance dataset, the `G3` column represents the final grade obtained by each student.

The final grades are analyzed using statistical methods and visualizations to understand their distribution. A histogram is used to display the frequency of different final grade values.

The analysis of final grades helps to understand how students performed in the examination and whether the grades are concentrated within particular ranges.

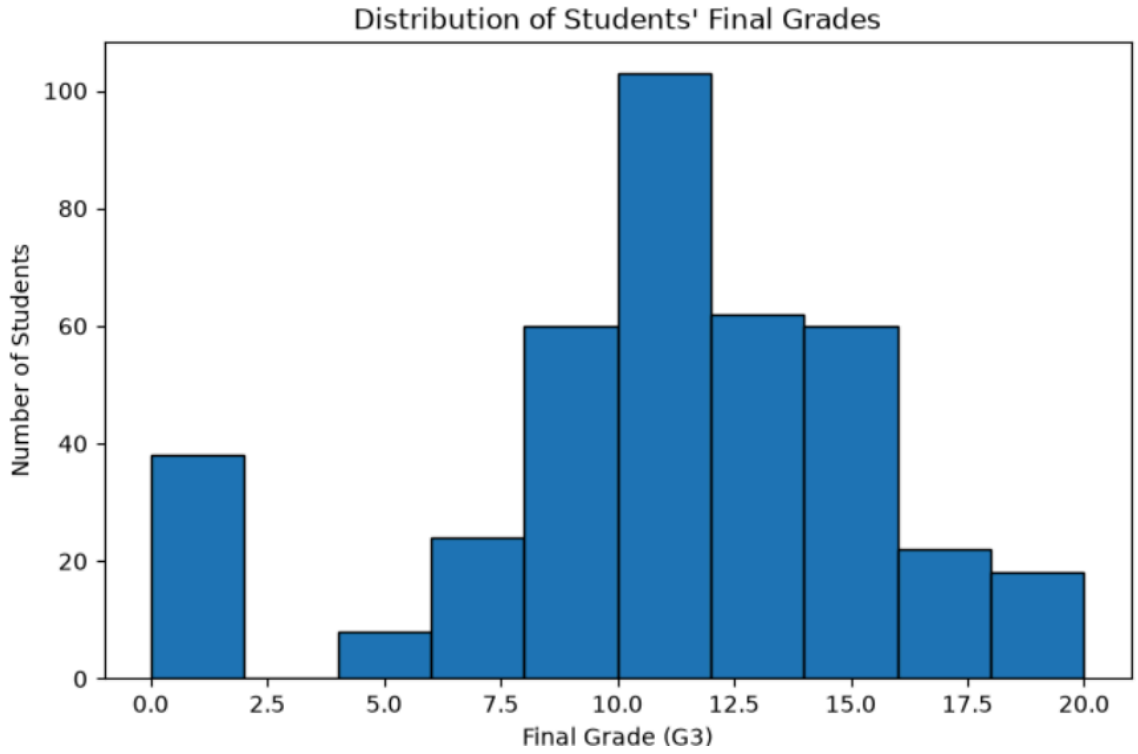
Fig 8: Distribution of students' final grades:

```
plt.figure(figsize=(8, 5))

plt.hist(df["G3"], bins=10, edgecolor="black")

plt.title("Distribution of Students' Final Grades")
plt.xlabel("Final Grade (G3)")
plt.ylabel("Number of Students")

plt.show()
```



12. ANALYSIS OF STUDY TIME AND ACADEMIC PERFORMANCE:

Study time is one of the academic factors examined in this project. The dataset contains a `studytime` attribute that represents the amount of time students spend studying each week.

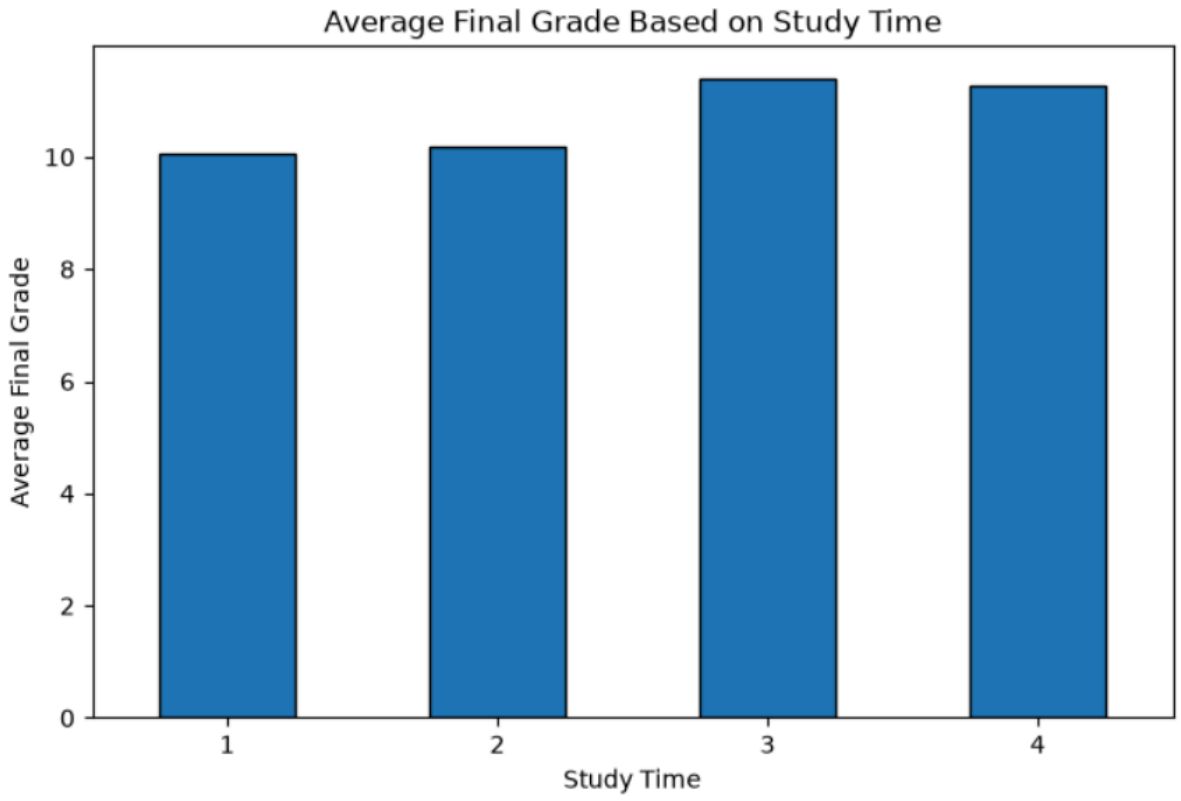
The study time attribute is compared with students' final grades (G3) using grouping and bar chart visualizations. This helps examine how academic performance varies across different study time categories.

The analysis provides a way to explore the association between weekly study time and students' final grades. However, study time alone cannot determine a student's academic performance.

Fig 9: Relationship between study time and average final grades:

```
study_grade.plot(kind="bar", figsize=(8, 5), edgecolor="black")

plt.title("Average Final Grade Based on Study Time")
plt.xlabel("Study Time")
plt.ylabel("Average Final Grade")
plt.xticks(rotation=0)
plt.show()
```



13. ANALYSIS OF ABSENCES AND ACADEMIC PERFORMANCE:

Attendance is another factor examined in the Student Performance dataset. The `absences` column represents the number of school days missed by each student.

A scatter plot is created using the number of absences on the X-axis and final grades (`G3`) on the Y-axis. Each point in the graph represents one student.

The visualization helps examine the relationship between school absences and final grades. It allows us to observe whether students with different attendance records show different academic performance patterns.

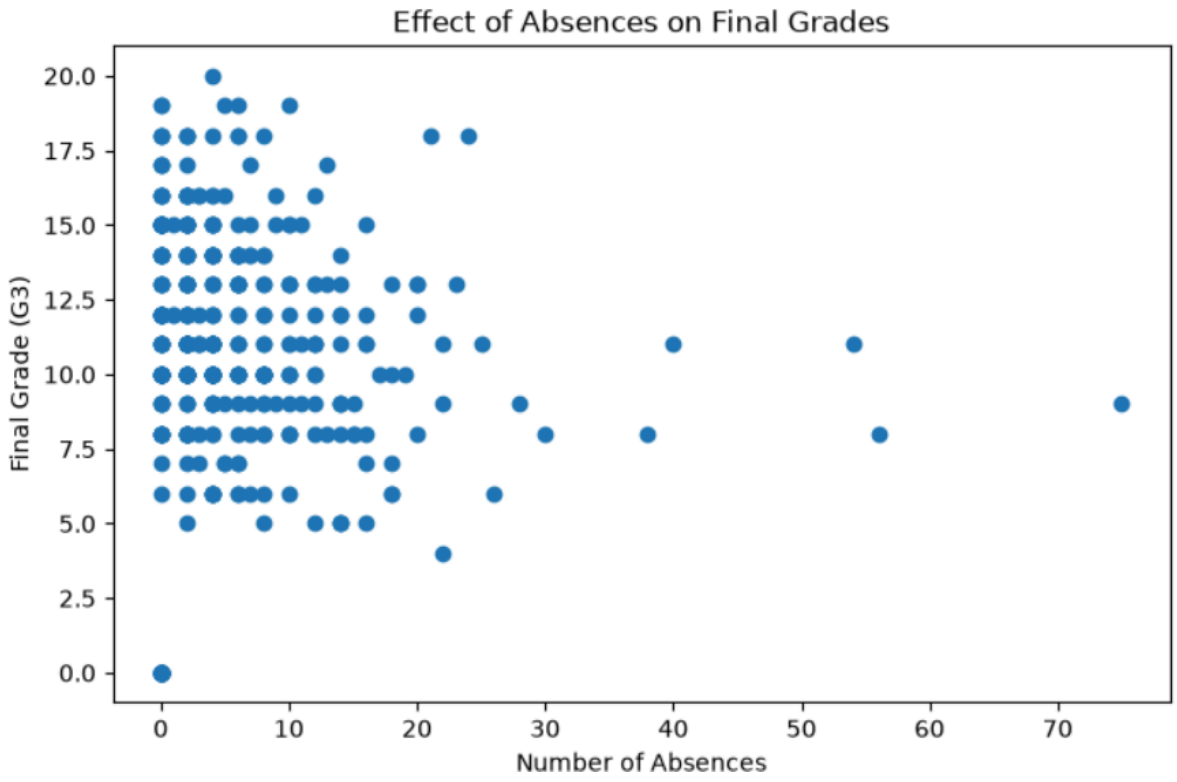
Fig 10: Relationship between student absences and final grades:

```
plt.figure(figsize=(8, 5))

plt.scatter(df["absences"], df["G3"])

plt.title("Effect of Absences on Final Grades")
plt.xlabel("Number of Absences")
plt.ylabel("Final Grade (G3)")

plt.show()
```



14. ANALYSIS OF PREVIOUS ACADEMIC FAILURES:

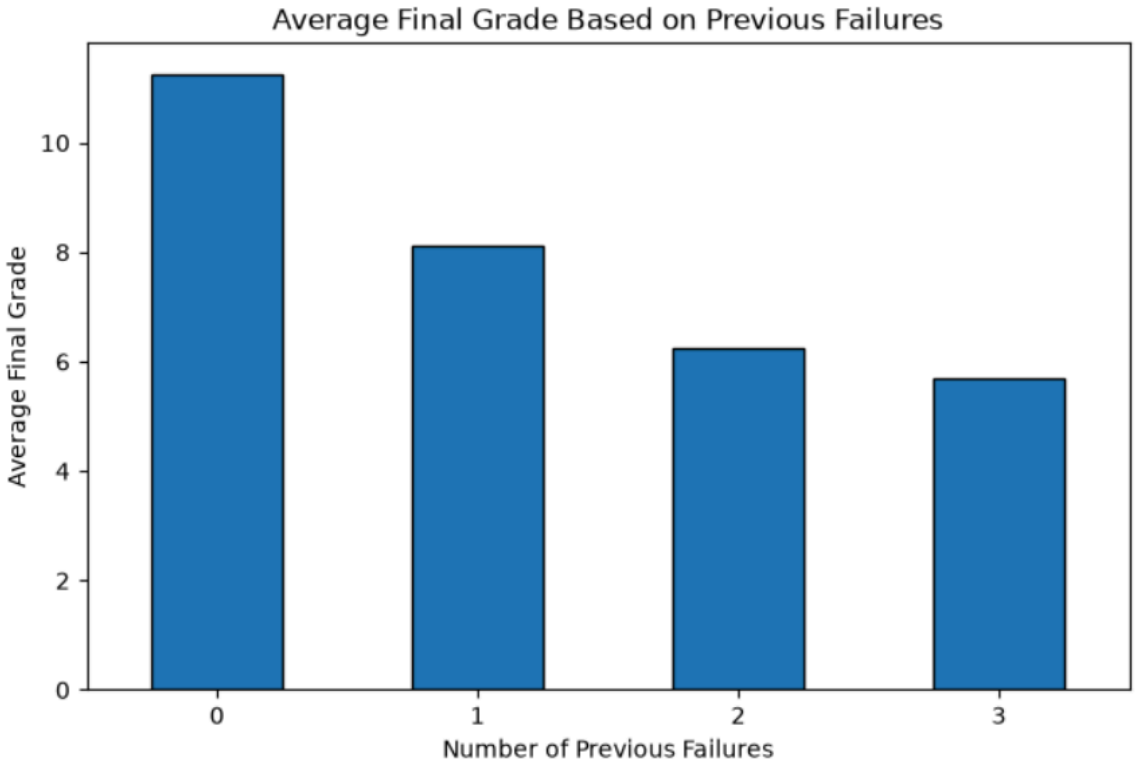
The `failures` attribute represents the number of previous class failures of each student. This factor is examined to understand its association with students' current academic performance.

The project uses grouping and visualization techniques to compare previous academic failures with final grades (`G3`). This analysis helps explore whether students with different academic histories show different patterns in their final grades.

Figure 11: Average final grades based on previous academic failures.

```
failure_grade.plot(kind="bar", figsize=(8, 5), edgecolor="black")

plt.title("Average Final Grade Based on Previous Failures")
plt.xlabel("Number of Previous Failures")
plt.ylabel("Average Final Grade")
plt.xticks(rotation=0)
plt.show()
```



15. ANALYSIS OF FAMILY SUPPORT AND SCHOOL SUPPORT:

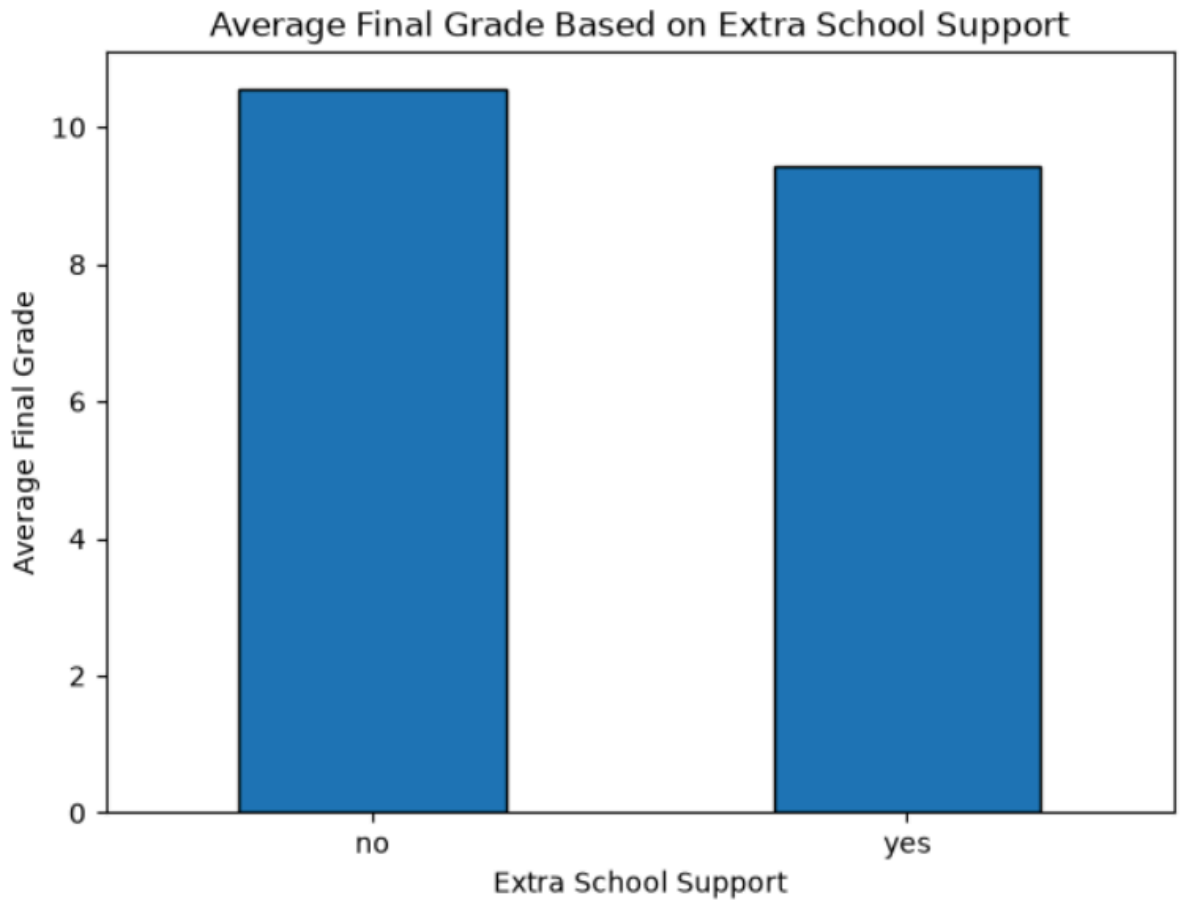
The `famsup` and `schoolsup` attributes represent the support received by students from their families and schools. These factors are examined to understand their association with students' academic performance.

The project uses grouping and visualization techniques to compare family support and school support with students' final grades (G3). This helps explore how academic performance varies among students receiving different levels of support.

Fig 12: Average final grades based on extra school support:

```
support_grade.plot(kind="bar", figsize=(7, 5), edgecolor="black")

plt.title("Average Final Grade Based on Extra School Support")
plt.xlabel("Extra School Support")
plt.ylabel("Average Final Grade")
plt.xticks(rotation=0)
plt.show()
```



16. ANALYSIS OF INTERNET ACCESS AND STUDENT PERFORMANCE:

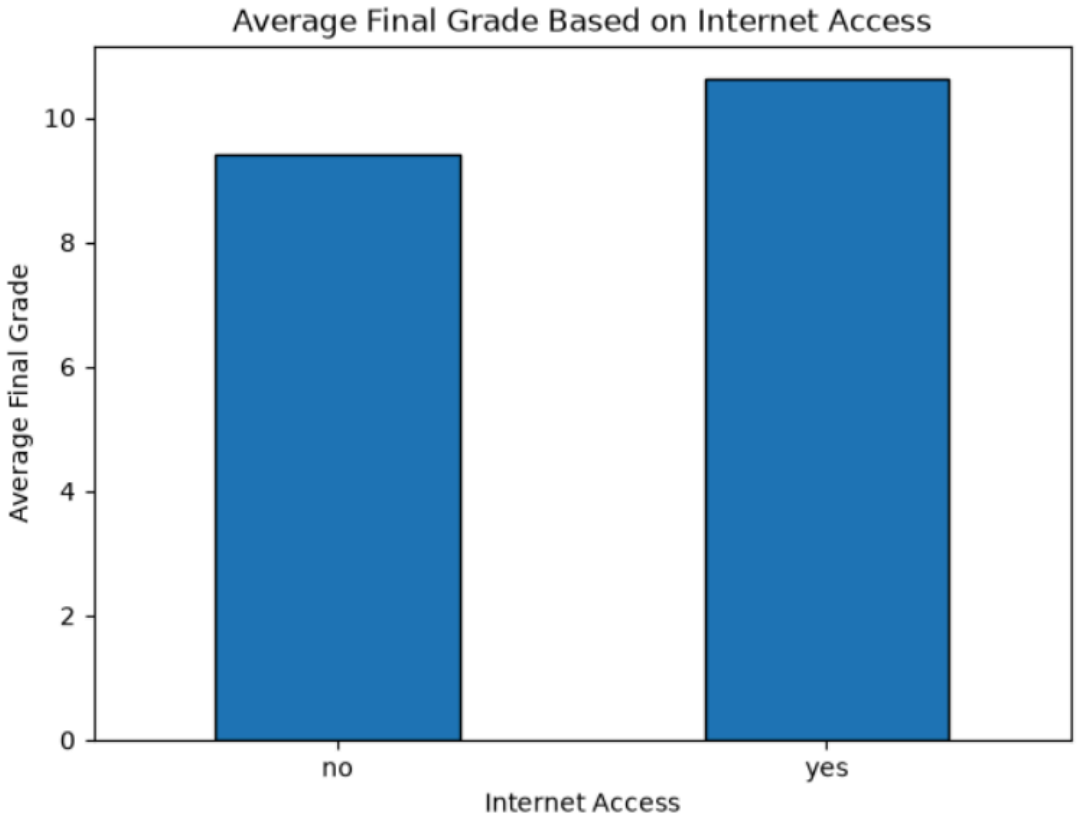
The `internet` attribute indicates whether a student has access to the internet at home. This factor is examined to explore its association with students' academic performance.

The project uses grouping and bar chart visualizations to compare students with and without internet access based on their final grades (`G3`). This helps identify patterns in academic performance across the two groups.

Fig 13: Average Final Grades Based on Internet Access:

```
internet_grade.plot(kind="bar", figsize=(7, 5), edgecolor="black")

plt.title("Average Final Grade Based on Internet Access")
plt.xlabel("Internet Access")
plt.ylabel("Average Final Grade")
plt.xticks(rotation=0)
plt.show()
```



17. ANALYSIS OF GENDER AND STUDENT PERFORMANCE:

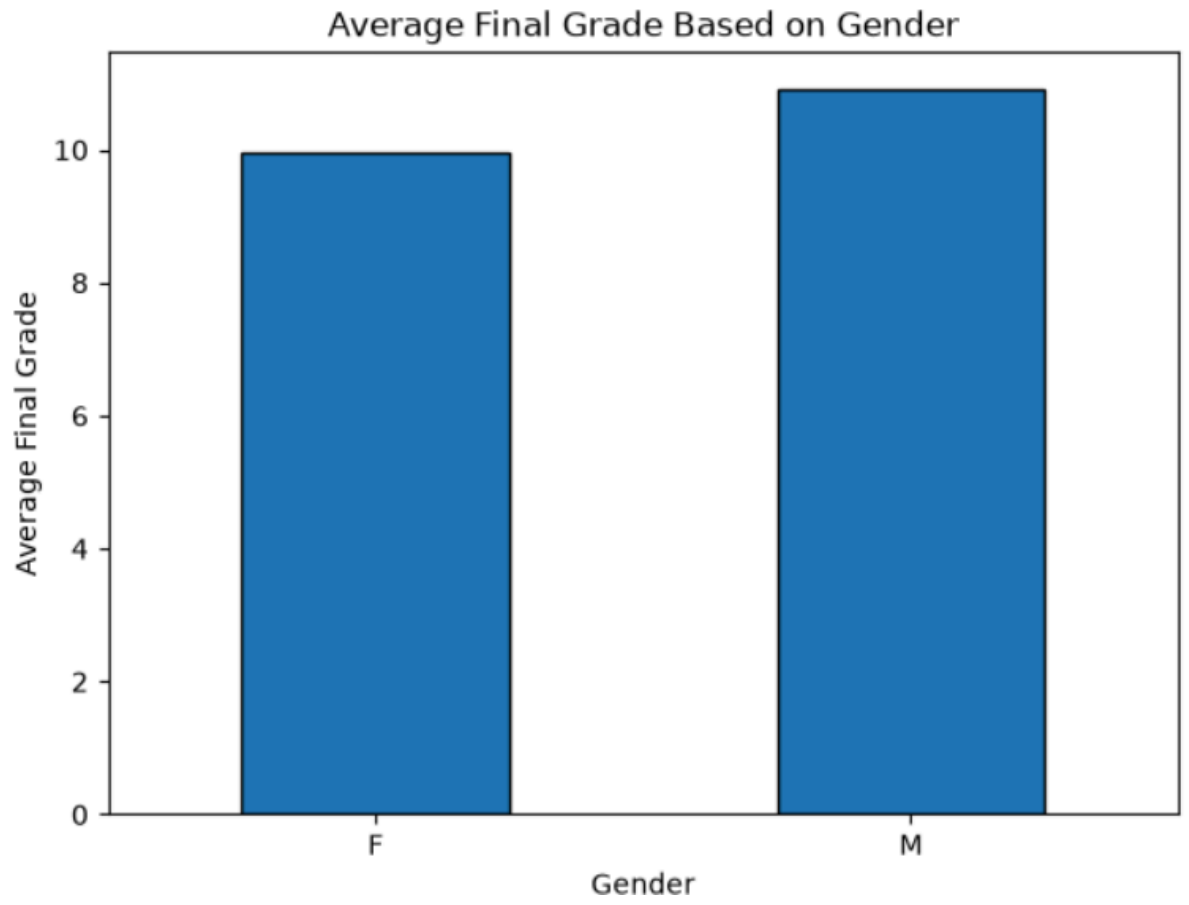
The `sex` attribute represents the gender of each student in the dataset. This attribute is examined to explore differences in academic performance between male and female students.

The project uses grouping and bar chart visualizations to compare the average final grades (G3) of male and female students. This helps identify patterns and differences in academic performance between the two groups.

Fig 14: Average Final Grades Based on Gender:

```
gender_grade.plot(kind="bar", figsize=(7, 5), edgecolor="black")

plt.title("Average Final Grade Based on Gender")
plt.xlabel("Gender")
plt.ylabel("Average Final Grade")
plt.xticks(rotation=0)
plt.show()
```



18. CORRELATION ANALYSIS:

Correlation analysis is used to examine the relationship between numerical variables in the Student Performance dataset. It helps identify how changes in one variable are associated with changes in another variable.

In this project, a correlation matrix is used to study the relationships between numerical attributes such as study time, absences, previous failures, and students' grades (`G1`, `G2`, and `G3`).

A heatmap is created using the Seaborn library to represent the correlation values visually. Different colours help identify positive, negative, and weak relationships between variables.

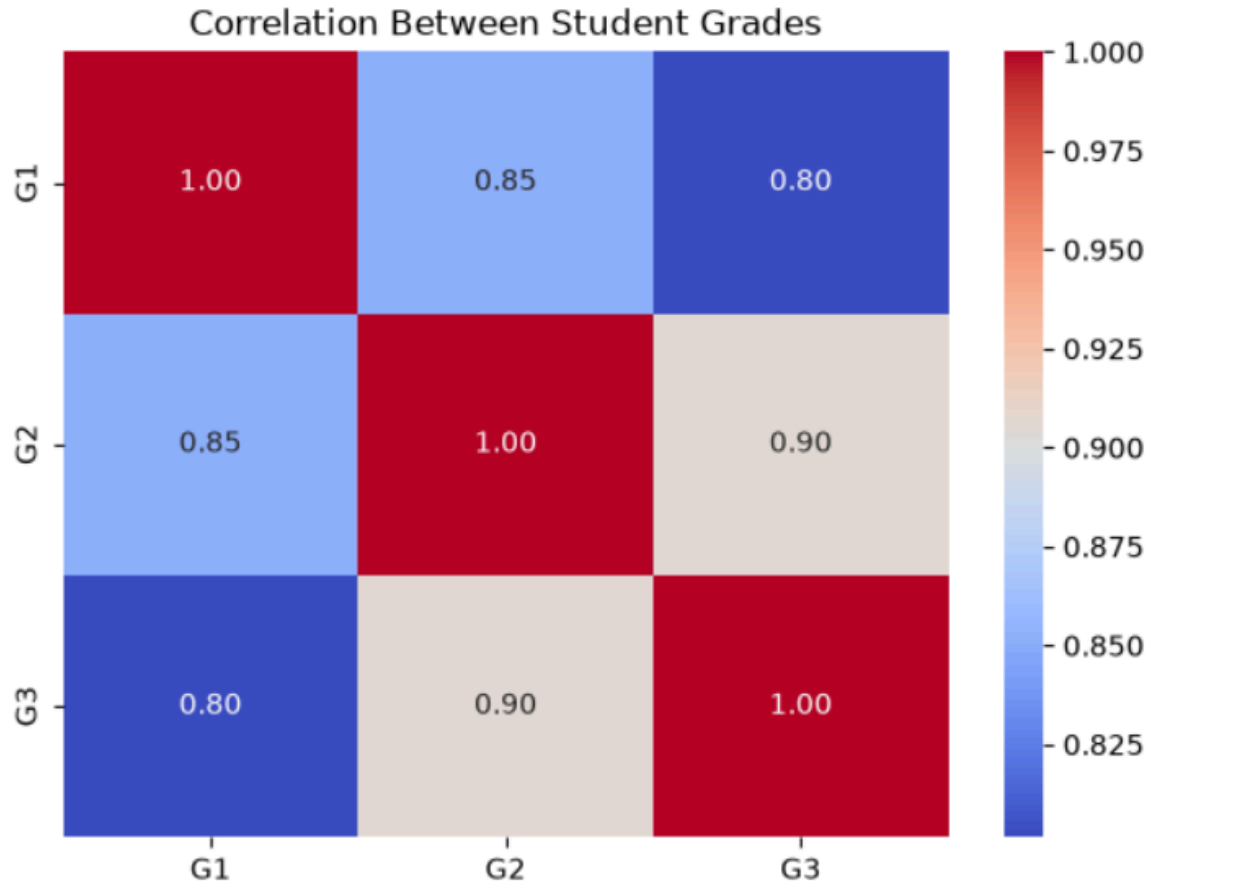
This analysis provides a better understanding of the relationships among different student attributes and their association with academic performance.

Fig 15: Correlation between first-period, second-period, and final grades (G1, G2 and G3):

```
plt.figure(figsize=(7, 5))

sns.heatmap(grade_correlation, annot=True, cmap="coolwarm", fmt=".2f")

plt.title("Correlation Between Student Grades")
plt.show()
```



19. PASS AND FAIL CLASSIFICATION:

In this project, students are classified into two categories based on their final grades (G3): pass and fail.

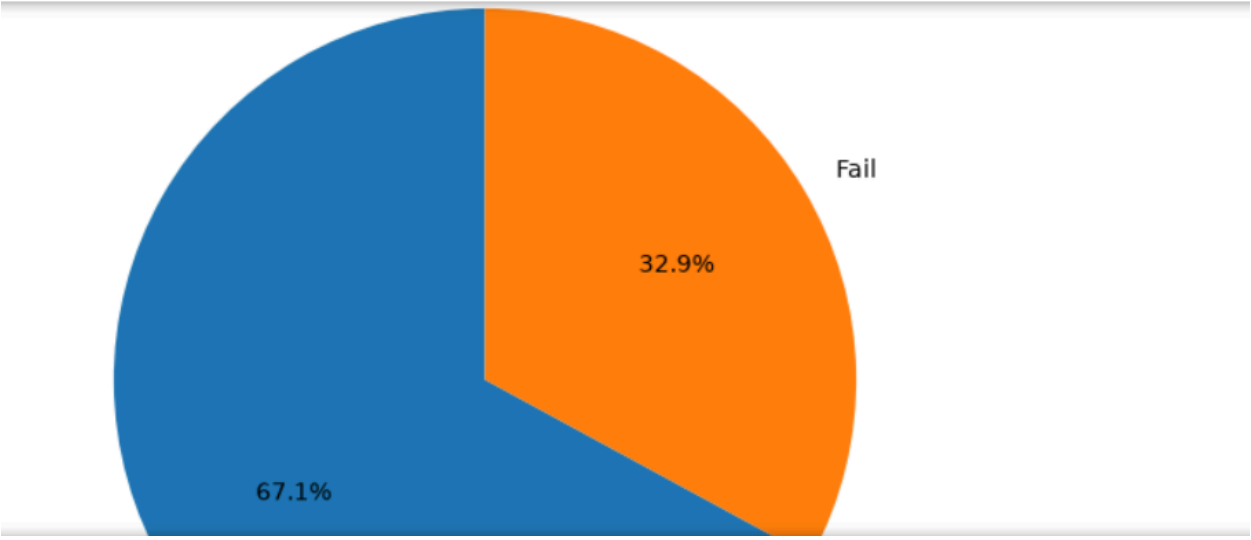
A new classification is created using the final grade. Students with a final grade of 10 or above are considered as pass, while students with a final grade below 10 are considered as fail.

A pie chart is used to visualize the distribution of students in the pass and fail categories. This visualization helps understand the overall academic performance of students in the dataset.

Fig 16: Pass and fail distribution of students:

```
result_count.plot(
    kind="pie",
    autopct="%1.1f%%",
    figsize=(7, 7),
    startangle=90
)

plt.title("Pass and Fail Distribution of Students")
plt.ylabel("")
plt.show()
```



20. OTHER STUDENT-RELATED FACTORS:

Apart from academic factors, the Student Performance dataset contains several personal and social attributes that can be explored through data analysis.

In this project, attributes such as age, free time, workday alcohol consumption, and other student-related characteristics are examined using statistical methods and visualizations.

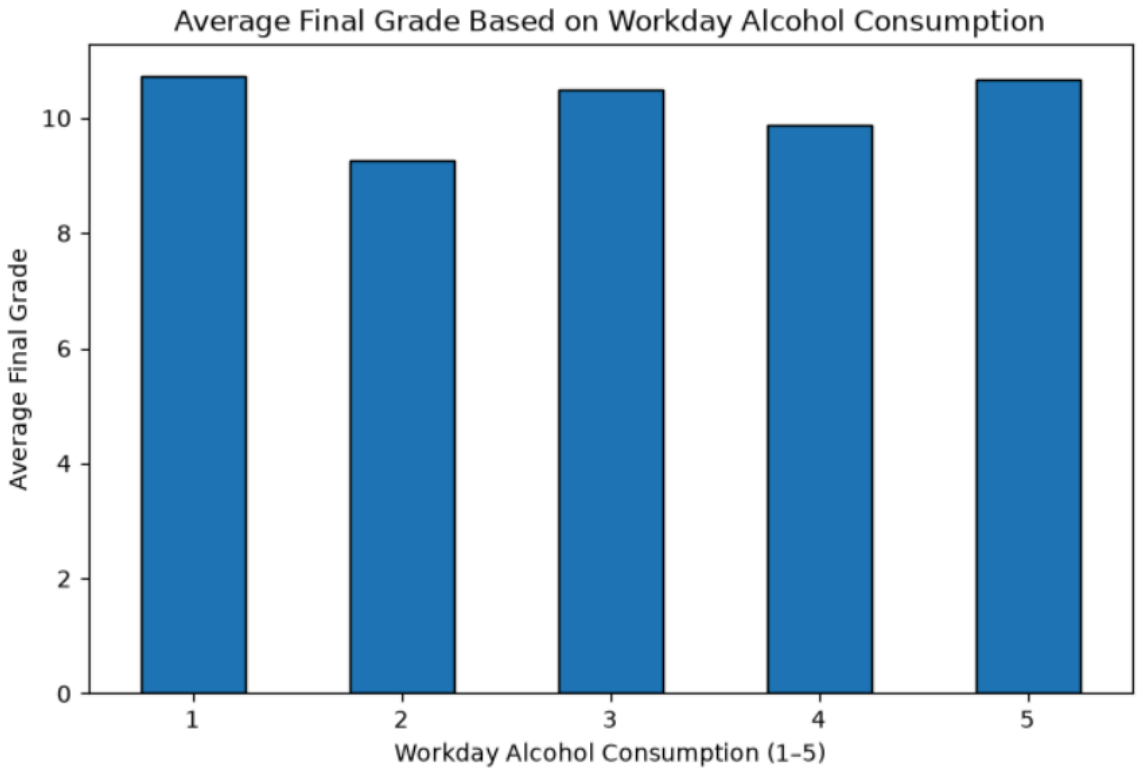
Different graphs, including bar charts and box plots, are used to explore these attributes and understand their patterns within the dataset.

This analysis provides a broader understanding of the students' personal and social characteristics and their association with academic performance.

Fig 17: Average final grades based on workday alcohol consumption:

```
alcohol_grade.plot(kind="bar", figsize=(8, 5), edgecolor="black")

plt.title("Average Final Grade Based on Workday Alcohol Consumption")
plt.xlabel("Workday Alcohol Consumption (1-5)")
plt.ylabel("Average Final Grade")
plt.xticks(rotation=0)
plt.show()
```



21. RESULTS AND OBSERVATIONS:

The Exploratory Data Analysis performed in this project provides useful insights into the Student Performance dataset.

The analysis and visualizations help examine the following observations:

1. The distribution of students' final grades (G3) can be understood using a histogram.

2. Study time can be compared with students' final grades to explore their relationship.
3. Absences and previous academic failures can be examined in relation to final grades.
4. Family support, school support, and internet access can be compared with academic performance.
5. Gender-based comparisons help explore differences in students' final grades.
6. Correlation analysis helps identify relationships among numerical attributes.
7. Pass and fail classification provides an overview of students' academic results.

Overall, the project demonstrates how Python-based data analysis and visualization techniques can be used to explore student performance and identify meaningful patterns.

22. CONCLUSION:

The Student Performance Analysis project was successfully carried out using Python and Exploratory Data Analysis (EDA) techniques. The Student Performance dataset was explored to understand students' academic, personal, and social characteristics.

Libraries such as Pandas, NumPy, Matplotlib, and Seaborn were used for data handling, statistical analysis, and visualization. Different graphs, including histograms, bar charts, scatter plots, box plots, pie charts, and heatmaps, helped examine the dataset and identify patterns.

The project explored the association of factors such as study time, absences, previous failures, family support, school support, internet access, and other student-related attributes with final grades.

Overall, this project provided practical experience in Python programming, data analysis, data visualization, and interpretation of results. It also helped develop a better understanding of how data analysis techniques can be applied to real-world educational datasets.

23. FUTURE ENHANCEMENTS:

This project can be further improved by applying additional data analysis and machine learning techniques.

Some possible future enhancements include:

1. Developing machine learning models to predict students' final grades.
2. Analyzing additional student-related factors to gain deeper insights.
3. Creating interactive dashboards for better data visualization.
4. Comparing academic performance across different subjects.
5. Using larger datasets to perform more detailed analysis.
6. Developing an early identification system to help recognize students who may need additional academic support.

These enhancements can make the project more useful for understanding student performance and supporting educational decision-making.

24. REFERENCES:

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3. NumPy Documentation – https://numpy.org/doc/

4. Matplotlib Documentation – https://matplotlib.org/stable/

5. Seaborn Documentation – https://seaborn.pydata.org/